

Monisha Badugu

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EDUCATION

Indian Institute of Information Technology, Gwalior

Madhya Pradesh, India

Bachelor of Technology in Computer Science Engineering | **CGPA: 8.11**

Expected May 2027

EXPERIENCE

Cyfrin

March 2026 – Present

Web3 Security Ambassador

Remote

- Selected from **2,000+** applicants as one of **19 students worldwide (top 1%)** for the Cyfrin Updraft program specializing in **smart contract security** and **Solidity vulnerability research**.
- Conducting technical research on **DeFi exploit patterns**, **reentrancy attacks**, and secure smart contract design; publishing educational security analyses and contributing to vulnerability bounty programs.

OPEN SOURCE

Hacktoberfest 2025

October 2025

Open Source Contributor

Remote

- Delivered **12+ merged pull requests** across open-source repositories, fixing production bugs, enhancing documentation, and improving automated test coverage using **JavaScript** and **Python**.
- Collaborated with global open-source maintainers via **Git-based workflows** to implement features, review code, and maintain production-ready code quality standards.

PROJECTS

LecSum | Python, OpenAI Whisper, Meta LLaMA 2, Gradio, NLP, Speech-to-Text

GitHub

- Developed an **AI-powered lecture summarization system** that converts recorded lectures into structured notes using **OpenAI Whisper ASR** and **Meta LLaMA 2**, automating transcription and reducing the need for manual note-taking
- Engineered a **multi-stage NLP pipeline** for audio ingestion, speech-to-text transcription, and abstractive summarization; deployed via a **Gradio web interface** with modular architecture for reproducible ML experimentation and deployment.

AquaRisk | Python, Scikit-learn, XGBoost, Random Forest, SVM, ANN, Streamlit, Fuzzy Logic

GitHub

- Built a **machine learning risk prediction framework** for groundwater fluoride contamination across India using **16,776+ environmental samples**, achieving **93% classification accuracy**.
- Implemented an ensemble of **Random Forest, XGBoost, SVM, and ANN** models with a **Mamdani Fuzzy Inference System** to generate interpretable risk scores; deployed an **interactive Streamlit dashboard** for geospatial risk visualization and model comparison.

Process Manager | C++, POSIX, Linux /proc, Makefile

GitHub

- Implemented a **Linux process monitoring tool** (htop-style) in **C++** parsing the **/proc filesystem** to track **100+ concurrent processes** with real-time CPU and memory usage metrics.
- Optimized performance with **POSIX system calls** and **UNIX signal handling**, achieving a **50ms refresh rate** while maintaining **< 2% CPU overhead** through efficient process scheduling and memory management.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, Bash, HTML, CSS

Frameworks & Libraries: React, Next.js, Node.js, Express.js

Databases: MongoDB, Supabase

Web & APIs: REST APIs, WebSockets

Blockchain: Solidity, EVM, Smart Contracts, Foundry, Web3

Tools: Git, Linux, Docker, Vercel, Postman

Coursework: Data Structures and Algorithms (DSA), Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN), Object-Oriented Programming (OOP), Computer Architecture (COA), Computer Security

ACHIEVEMENTS

- Awarded the **NTSE Scholarship** by NCERT, ranking in the **top 0.1% nationally** among **1.5M+ candidates**.
- Claimed **1st Place** at **LNM Hacks 8.0** (by Monad), outperforming **450+ teams** across **160+ projects**.
- Winner of **Monad Blitz Delhi**, a competitive **6-hour Web3 sprint hackathon**.
- Google WeCode Finalist** — selected through competitive technical evaluation in a highly selective applicant pool.
- Finalist, D.E. Shaw Ascend Educare** — shortlisted for exceptional analytical and technical aptitude.
- Ranked **266 / 5,725** in **CodeClash 2025**, a national-level competitive programming contest.
- Semi-Finalist, Tata Imagination Challenge 2025** — selected for innovative problem-solving skills.
- Reached among the **top 17 teams** out of **736+ registrations** at **HACKSTREET 3.0**.
- Served as **Student Council Executive** at **Infotsav**, coordinating logistics for **1,000+ participants**.
- Core member of the **AASF Technical Club** (college tech club); organized **30+ technical events, workshops, and hackathons**, increasing participation by **40%**.